

A Case Report of Overlapping Miller Fisher Syndrome, Guillain-Barré Syndrome, and the Bickerstaff Brainstem Encephalitis

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Introduction: Patients with overlapping Miller Fisher syndrome (MFS), Guillain-Barré syndrome (GBS), and the Bickerstaff brainstem encephalitis (BBE) have rarely been reported, and the diverse clinical features and sequelae are challenging for physicians. Through this case report, we present the clinical course, treatment, recovery, and sequelae of an overlapping MFS, GBS, and BBE syndrome and provide a review of the literature.

Case Report: A 76-year-old woman was initially presented with diplopia, ophthalmoplegia, and ataxia, but she later developed weakness of limbs, respiratory failure, deterioration of consciousness, and cognitive impairment. After IV immunoglobulin and corticosteroid therapy, she began to recover consciousness and from the weakness from day 18. After 8 months of rehabilitative therapy, she was able to swallow food without difficulty, but she still cannot walk without assistance, showed cognitive impairment, and the tracheostomy remained in place because of persistent respiratory muscle weakness.

Conclusions: As the clinical course of MFS-GBS-BBE overlapping syndrome is variable, involving not only the peripheral but also the central nervous system, the establishment of a comprehensive therapeutic strategy is warranted.

Key Words: Miller Fisher syndrome, Guillain-Barré syndrome, central nervous system, brainstem, encephalitis

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There are several clinical subtypes of Guillain-Barré syndrome (GBS) and its variants Miller Fisher syndrome (MFS) and the Bickerstaff brainstem encephalitis (BBE) with different neurological features and presentations.^{1,2} Although the typical clinical features of GBS, MFS, and BBE are well-recognized, GBS is also used as a broad term to describe a number of related acute autoimmune neuropathies and it sometimes cannot comprehensively describe the full spectrum of related diseases including overlapping cases.^{1,3} The evidence from overlapping cases supports the classification of MFS, GBS, and BBE as subtypes within a clinical spectrum of the

same disease, possibly affecting the peripheral and/or central nervous systems.^{1,3,4}

Patients with overlapping MFS, GBS, and BBE have rarely been reported, and the clinical course and prognosis of the disease have not been described in detail. Herein, we report a Korean female with a GBS overlapping syndrome showing the manifestations of peripheral and central nervous system involvement. The clinical diagnosis of GBS was supported by cerebrospinal fluid (CSF) albuminocytological dissociation, serologic anti-GQ1b antibodies, and an electrophysiological evidence of acute motor and sensory axonal neuropathy. This case report illustrates the challenging presentation, treatment, recovery, and sequelae of an overlapping MFS-GBS-BBE syndrome and provides a review of the literature.

CASE REPORT

A 76-year-old Asian woman developed myalgia, fever, nausea, and vomiting for 3 days. She had been diagnosed with type 2 diabetes mellitus and hypertension 10 years before. She visited the emergency room due to a gradual aggravation of her symptoms and the development of diplopia and ophthalmoplegia (day 1). At the emergency room, her mental status was alert without limb weakness. She had gait disturbance with an ataxic pattern. Her deep tendon reflex was hypoactive, and no pathologic reflex was observed. She was admitted to the neurological department with a diagnosis of MFS. Her CSF on day 1 showed increased protein levels (94.8 mg/dL) and 5 white blood cells/mm³. On day 2, she could not open her eyes spontaneously due to severe ptosis. A brain magnetic resonance imaging on day 3 showed no remarkable findings. Dysarthria and dysphagia were newly developed, and a Levin tube was inserted. She showed muscle weakness of Medical Research Council (MRC) grade 4 in the upper and lower extremities. On day 3, her muscle power was aggravated to MRC grade 3 in all extremities. Dyspnea developed on day 4 and her oxygen saturation dropped to 82%. As her saturation level was not maintained, she was intubated and assisted ventilation was started. The arterial CO₂ levels remained constant within the normal range. On day 5, her limb weakness further progressed to MRC grade 1. Her mental status decreased from alert to a stupor state, and she showed a sluggish pupil reflex. She was positive for immunoglobulin (Ig) G against GQ1b (1:1600), but negative for IgGs against GM1, GD1b, GT1a, or GT1b (normal range for each antibody <1:40) in a serum sample obtained on day 2. Electromyography and nerve conduction studies revealed acute motor and sensory axonal neuropathy and bilateral facial neuropathy.

A clinical diagnosis of GBS was considered and she received IV immunoglobulin (IVIG) therapy on day 6 for 5 days (400 mg/kg/d). However, IVIG administration had no immediate effect on her motor weakness and deteriorated mental status. A follow-up brain magnetic resonance imaging on day 13 revealed no abnormalities in the brainstem and cerebral hemispheres. A follow-up CSF examination on day 16 showed more increased protein levels (401.9 mg/dL) and 0 white blood cells. Electroencephalography on day 17 showed diffuse cerebral dysfunction without epileptiform discharges. As her limb weakness and stupor mental status were persistent after IVIG treatment, on day 17, the patient received additional therapy with 1000 mg/d methylprednisolone

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for 5 consecutive days. Her mental status began to recover to an alert state from day 18 with a tendency to wax and wane. Her limb weakness improved to MRC grade 3 on day 21; however, ventilator weaning was delayed due to the impaired ventilation and copious secretions. A tracheostomy was performed on day 23 and mechanical ventilation was removed on day 24. After recovery of consciousness, she showed a cognitive decline compared with before the onset of the disease.

During the hospital days, rehabilitative strengthening exercises, respiratory exercises, balance training, and occupational therapy were gradually applied. She began to open her eyelids and move eyeballs spontaneously on day 45. There was a large amount of aspiration during a videofluoroscopic study on day 48, which was caused by decreased pharyngeal contraction; therefore, we decided to continue the Levin tube feeding. She could sit on the side of the bed on day 53. After discharge from our hospital (day 54), she underwent an outpatient daily rehabilitation program. On day 69, she exhibited a normal swallowing function and started a regular diet without difficulty. On day 150, her muscle power was improved to MRC grade 4 in some proximal muscles and she could stand unassisted for a short time. The peak cough flow on day 194 was 50 m/s due to persistent respiratory muscle weakness and attempts to remove the tracheostomy failed. On day 247, she could not ambulate independently, but she could walk indoors for a short distance with the aid of a walker.

DISCUSSION

This case demonstrates the variable and unpredictable course of overlapping MFS, GBS, and BBE. To our knowledge, 6 cases of overlapping MFS-GBS-BBE, including the current patient, have been reported since 2002 (age, 20 to 76 y; M:F, 2:4), as shown in Table 1.⁵⁻⁹ An unsteady gait and oculomotor dysfunction are commonly observed as the initial presentation. The overlap of the second entity occurred within 2 weeks in all reported cases, and the overlap of the third entity varied from day 5 to day 42. The sequence of the entities differed among the reported cases. Albuminocytological dissociation in the CSF was detected in 4 cases (days 1 to 21). Axonal neuropathy was observed in 4 cases, which is consistent with previous reports that axonal rather than demyelinating forms of GBS overlap with MFS.¹⁰ Only 1 case reported by Stevenson et al⁶ showed demyelinating neuropathy, which exhibited the best prognosis of independent ambulation on day 27. Anti-GQ1b antibodies are thought to cause ophthalmoplegia and cerebellar-type ataxia.¹¹ It is assumed that the impaired consciousness in BBE is also mediated by the expression of GQ1b ganglioside in the reticular formation.¹ Consistent with our review, anti-Gq1b antibodies were detected in 4 of 6 cases.⁵⁻⁹ The clinical features of our patient according to the new diagnostic classification suggested by the GBS classification group are summarized in Figure 1.¹ We concluded our patient to have BBE by reviewing the diagnostic criteria suggested by Koga et al.¹² BBE is characterized by the presence of anti-GQ1b antibodies and a typical clinical picture of acute progressive external ophthalmoplegia, ataxia, and impaired consciousness by 4 weeks, followed by spontaneous recovery within 12 weeks after onset. The patient met the diagnostic criteria for BBE except for uncertain recovery of ataxia within 12 weeks, which is due to the severe weakness by overlapping of GBS.

IVIg or plasma exchange (PE) monotherapy are recommended treatments for GBS. The combined treatment with IVIg, PE, steroids, or immunoadsorption yielded no significant difference in patient outcome compared with monotherapy.² However, there are no established treatment regimens or clinical trials for overlapping cases. As shown in Table 1, all of the reported MFS-GBS-BBE overlapping cases were treated differently. After the initial treatment with IVIg or PE, additional treatments such as multiple sessions of IVIg or PE, steroids, and immunoadsorption were used in 4 cases. The deteriorating

TABLE 1. Reported Cases of Overlapping Miller Fisher Syndrome, Guillain-Barré Syndrome, and the Bickerstaff Brainstem Encephalitis

Age/Sex	Initial Symptoms	Comorbidity	Clinical Course	Albuminocytological dissociation (day)	NCS	Antibodies (Sampling Day)	Treatment	Outcome
Arai et al ⁵	34/F	Diplopia, unsteady gait	Not reported	BBE day 1 MFS day 2 GBS day 5	Not reported	GQ1b (day 2)	PE (7 sessions)	Tandem gait on day 22
Stevenson et al ⁶	43/F	Facial numbness, diplopia, vertigo, extremity tingling sense, and weakness	Not reported	BBE day 1 GBS day 9 MFS day 10	Demyelinating	GM1 (not reported)	IVIg (1 session)	Independent gait on day 27
Fujii et al ⁷	72/F	Diplopia, unsteady gait	Not reported	MFS day 1 GBS day 3 BBE day 5	AMSAN	GM1, GD1a, GQ1b, GT1a (day 6)	IVIg (2 sessions) Immunoadsorption	Full recover after 10 mo
Han et al ⁸	20/M	Unsteady gait	Asthma	MFS day 1 GBS day 12 BBE day 21	AMSAN	Negative	IV Dexamethasone	Walk around with support on day 132
Pegg et al ⁹	56/M	Hands tingling, unsteadiness, diplopia	Negative	MFS day 1 GBS day 7 BBE day 42	AMSAN	GD1a, GQ1b (not reported)	IVIg (2 sessions) PE	Walk with a frame after 4-5 mo
Our case 2018	76/F	Diplopia, ophthalmoplegia, ataxia	Diabetes HTN	MFS day 1 GBS day 3 BBE day 5	AMSAN	GQ1b (day 2)	IVIg (1 session) IV Methylprednisolone	Stand with assistance for a short time after 6 mo

AMSAN indicates acute motor sensory axonal neuropathy; HTN, hypertension; IVIg, IV immunoglobulin; NCS, nerve conduction study; PE, plasma exchange.

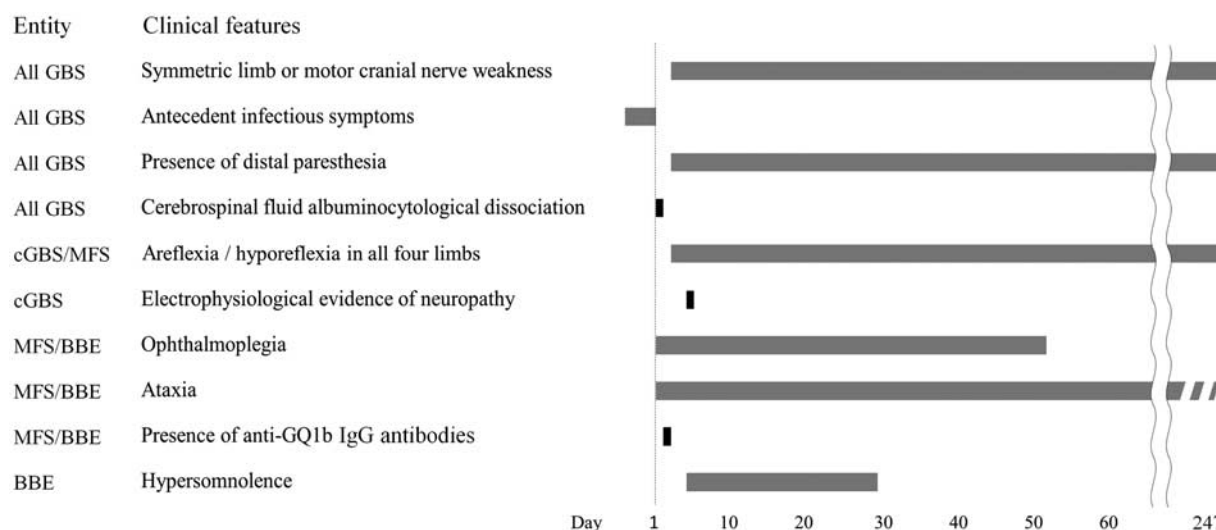


FIGURE 1. The clinical course of the current patient showing all core clinical features of GBS, MFS, and BBE. The timing of recovery of ataxia is uncertain. BBE indicates Bickerstaff's brainstem encephalitis; cGBS, classic Guillain-Barré syndrome; MFS, Miller Fisher syndrome.

course of overlap syndrome may result in inconsistent treatments. After 6 months, most patients with GBS and MFS show independent ambulation, and 66% of BBE patients achieve complete remission.^{4,13} However, the prognosis of MFS-GBS-BBE overlapping cases has not yet been identified. Most patients in previous case reports showed a favorable outcome at 6 months. However, after 8 months, the current patient, who had underlying diabetic polyneuropathy and is the oldest reported case to date, could not walk independently and the tracheostomy was maintained. The most difficult sequelae remaining in our patient are the weakness in the respiratory muscles and cognitive impairment. Considering the normal arterial CO₂ levels, her respiratory failure seems to be caused by peripheral and cranial neuropathy rather than brainstem dysfunction. Cognitive decline was the least predicted sequela and is thought to be an evidence of the involvement of central nervous system. Rehabilitative training is continuing with an emphasis on respiratory muscle exercise and cognitive training. Multicenter studies are needed to identify appropriate treatment strategies and predict the prognosis of overlapping cases.

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